

Omics Era

With the advent of Omics Era people start to be able to answer old big biological question:

Ex: Human genome

But also thinks to new questions:

Explore microbiome world, compare human genomes, find new genes and functions, etc...

A word cloud of bioinformatics and genomics terms. The words are arranged in a roughly circular shape, with 'genomics' being the largest and most central. Other prominent words include 'personalized medicine', 'computational biology', 'systems biology', 'metagenomics', 'cancer', 'genomics', 'exome analysis', 'machine-learning', 'pathway analysis', 'database-web applications', 'translational informatics', 'algorithm development', 'RNA-seq', 'protein interaction networks', 'bioinformatics software', 'Microarrays', 'sequence alignment', 'methyl-seq', 'integrative analysis', 'health informatics', 'domain-domain interactions', 'enzyme classification', 'ChIP-seq', 'variant-calling', 'protein modeling', 'clinical informatics', 'functional annotation', 'phylogenetics', and 'methyl-seq'. The colors of the words range from dark red to yellow.

pathway analysis
machine-learning
cancer
genomics
exome analysis
computational biology
personalized medicine
database-web applications
systems biology
metagenomics
algorithm development
RNA-seq
protein interaction networks
bioinformatics software
Microarrays
sequence alignment
methyl-seq
integrative analysis
health informatics
domain-domain interactions
enzyme classification
ChIP-seq
variant-calling
protein modeling
clinical informatics
functional annotation
phylogenetics

Examples of different types of omics include:

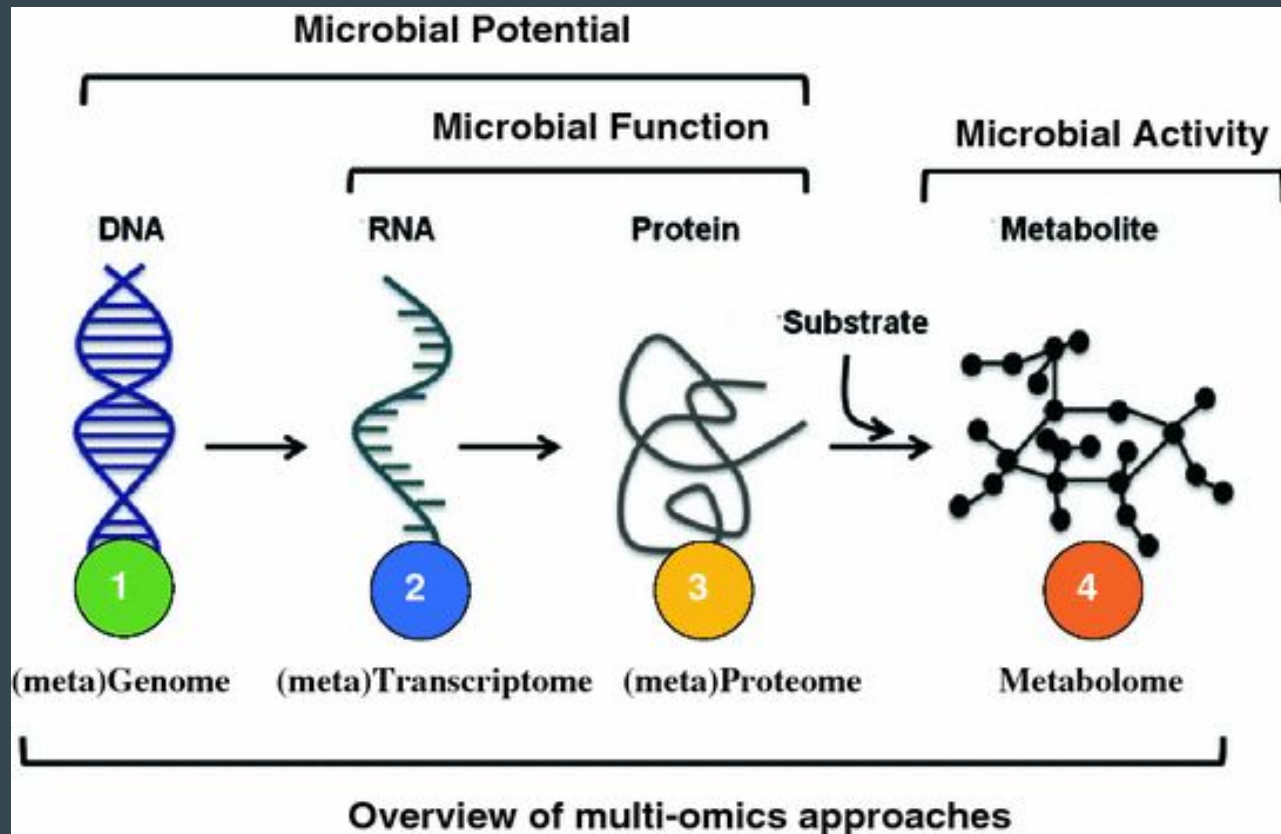
Genomics: studying genomes - entire set of genes, "the genetic landscape" of an organism

Transcriptomics: studying transcriptomes - the sets of all RNA molecules, including mRNA, rRNA, tRNA, and other non-coding RNAs

Proteomics: studying proteomes - entire sets of proteins produced by an organism

Metabolomics: metabolomics is the "systematic study of the unique chemical fingerprints that specific cellular processes leave behind", the study of their small-molecule metabolite profiles

Pharmacogenomics: the effect of variations within the human genome on response to drugs



Before Omics Era

1



Unknown
organism

>

2



DNA
extraction

>

3



Amplification
of target
DNA

>

4

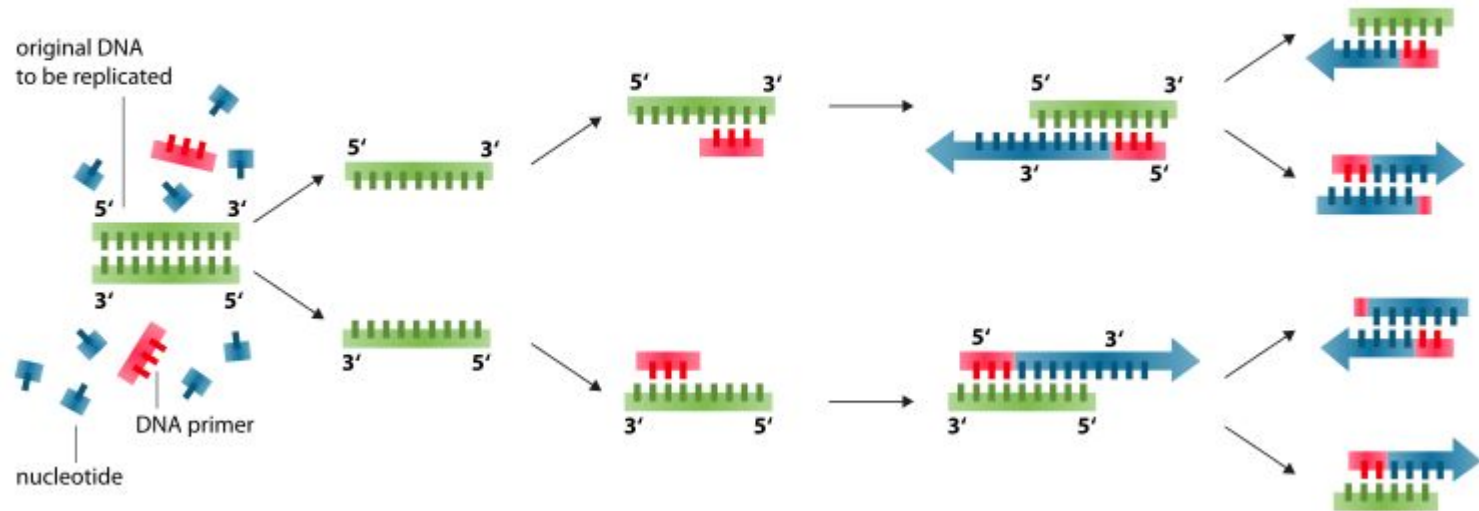
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Sanger
Sequencing

N. antiopa

Match with Ref
Database and
identification

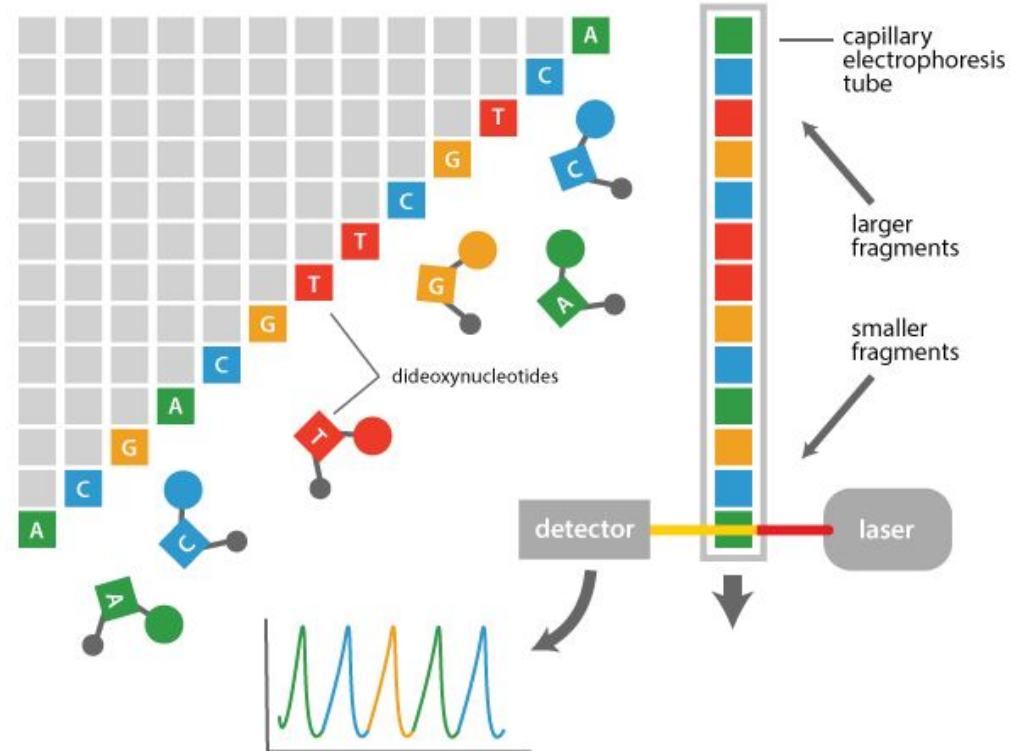
Polymerase chain reaction - PCR



Sanger sequencing



Sanger Sequencing



Welcome in Omics Era

1

>

2

>

3

>

4



Unknown
organisms -
eDNA



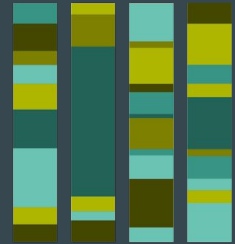
eDNA
extraction



Amplification
of target
DNA

tttgagtatacaact
ttcgagcatatcgact
aacgtccaaaggagt
ttggagcatatcgact
aaggtccaaagagt
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atcgtccaatggagt
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aacgtccaaaggagt
tttgagtatacaact

High-Throughput
Sequencing

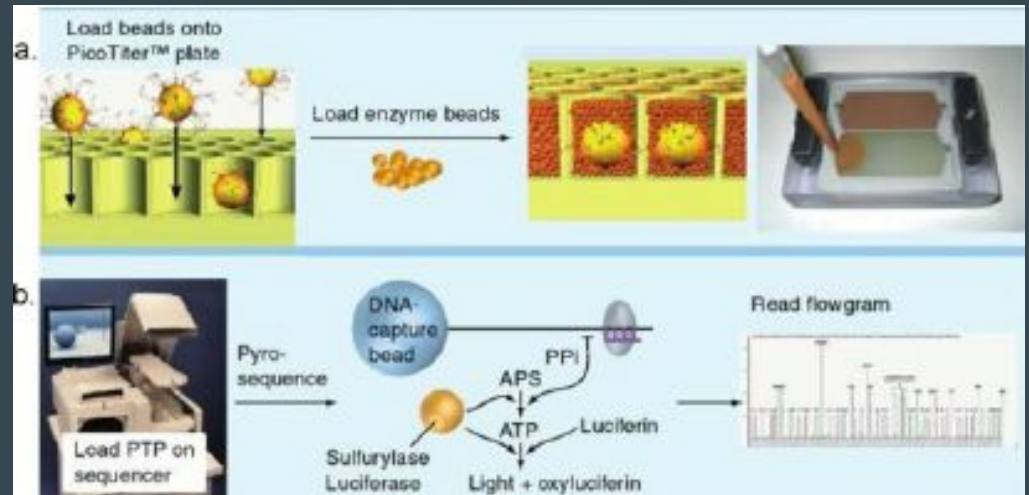


Match with Ref
Database and
identification

Historical Background

In 2005 the Roche 454 sequencing technology was the first NGS platform released on the market.

This was the breakthrough leading to the advent of metagenomic approach.



HTS - High-Throughput Sequencing Technologies



HiSeq2000/2500



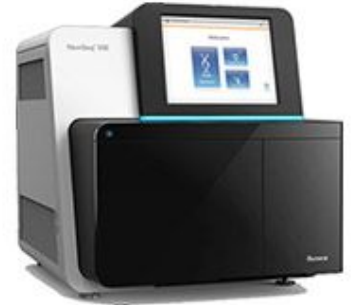
Ion PGM



Ion Proton



Pacbio RS II

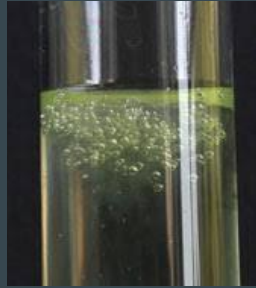


NextSeq500

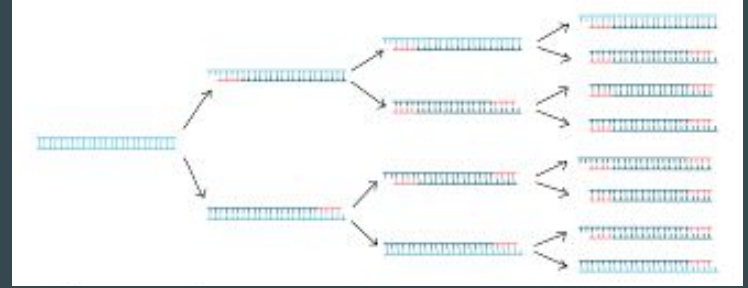
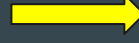
Workflow example



Sampling



Metagenome Extraction



Marker amplification &
library preparation

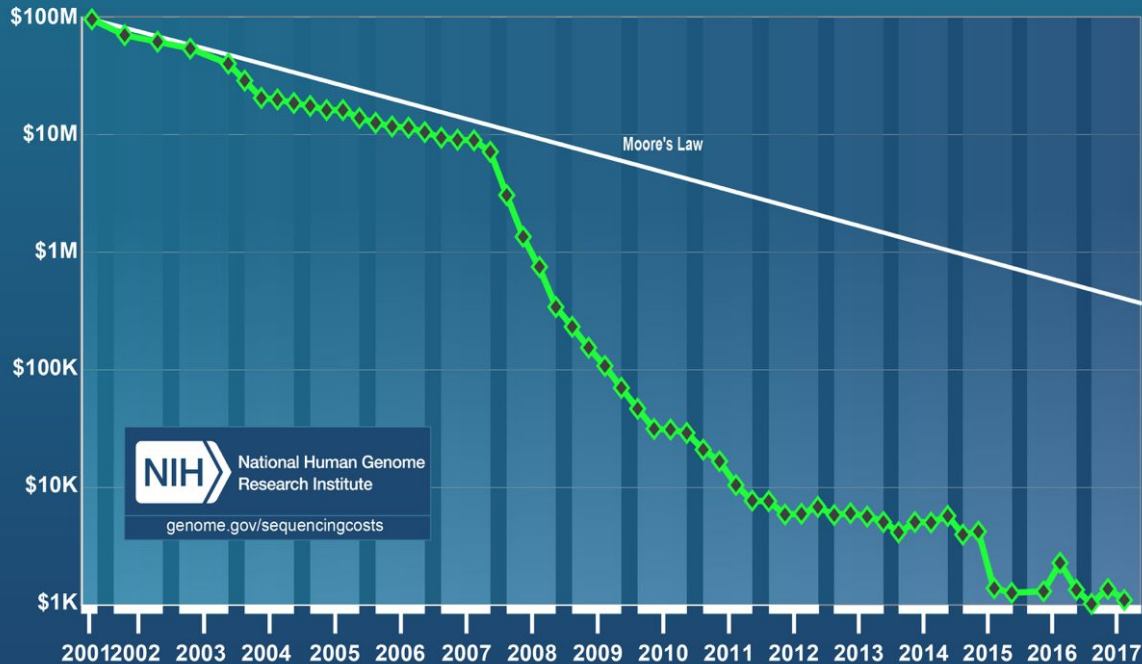


NGS Sequencing



Bioinformatic Analysis

Cost per Genome



In the last 10 years the sequencing cost dropped down.

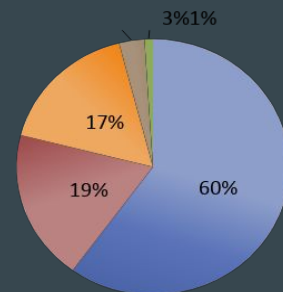
On Y axis a logarithmic scale of sequencing costs per Mb of DNA sequence (in dollars).

<https://www.genome.gov/27541954/dna-sequencing-costs-data/>

Omics market

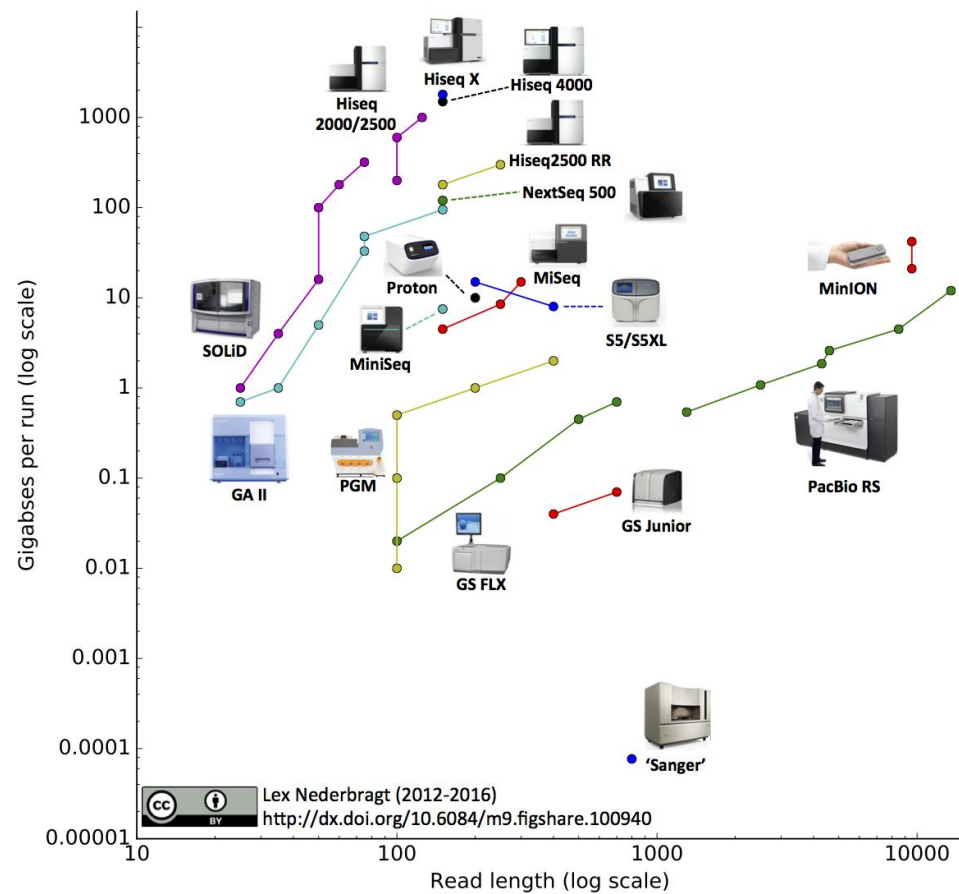


NGS Market



■ Illumina ■ SOLID ■ Roche 454 ■ Ion Torrent ■ PacBio

>7389 total machines situated in 1027 centres. Source: omicsmap.com



HTS field guide

Instrument	Amplification	Run time	Millions of Reads/run	Bases / read	Reagent Cost/run	Reagent Cost/Gb	Reagent Cost/Mread	bp/run	Gbp/run	cost/Gb
Sanger - Applied Biosystems 3730 (capillary)	PCR, cloning	2 hrs.	0.000096	650	\$144	\$2,307,692.31	\$1,500,000.00	62,400	0.00	\$2,307,692.31
Illumina MiSeq v3	bridgePCR	56 hrs.	25	600	\$1,530	\$102.00	\$61.20	15,000,000,000	15.00	\$102.00
Illumina HiSeq 2500 - high output v4	BridgePCR	40 hrs.	2000	50	\$6,240	\$62.40	\$3.12	100,000,000,000	100.00	\$62.40
Illumina HiSeq 2500 - high output v4	BridgePCR	6 days	2000	250	\$14,410	\$28.82	\$7.21	500,000,000,000	500.00	\$28.82
Ion Torrent – PGM 318 chip v2	emPCR	7 hrs.	5.5	400	\$874	\$397.27	\$158.91	2,200,000,000	2.20	\$397.27
Oxford Nanopore MinION (std. speed; low volume user)	None - SMS	varies	0.6	10000	\$900	\$150.00	\$1,500.00	6,000,000,000	6.00	\$150.00
Pacific Biosciences RS II	None - SMS	≤6 hrs.	0.055	12000	\$200	\$303.03	\$3,636.36	660,000,000	0.66	\$303.03

<http://www.molecularrecologist.com/next-gen-fieldguide-2016/>

Illumina MiniSeq, MiSeq, NextSeq, HiSeq

Illumina is the current undisputed leader in the high-throughput sequencing market. Illumina currently offers sequencers that cover the full range of data output. More details are available on the Illumina sequencers page.

Up to 300 million reads (HiSeq 2500)

Up to 1500 GB per run (GB = 1 billion bases)

IonTorrent PGM, Proton

The IonTorrent targets more specialized clinical applications.

Up to 400bp long reads

Up to 12 GB per run

PacBio Sequel

This is the leader in long read sequencing. More details on the PacBio sequencers page.

Up to 12,000 bp long paired end reads

Up to 4 GB per run



MinION

A portable, miniaturized device that is not yet quite as robust and reliable as the other options. More details on the MinION sequencers page.

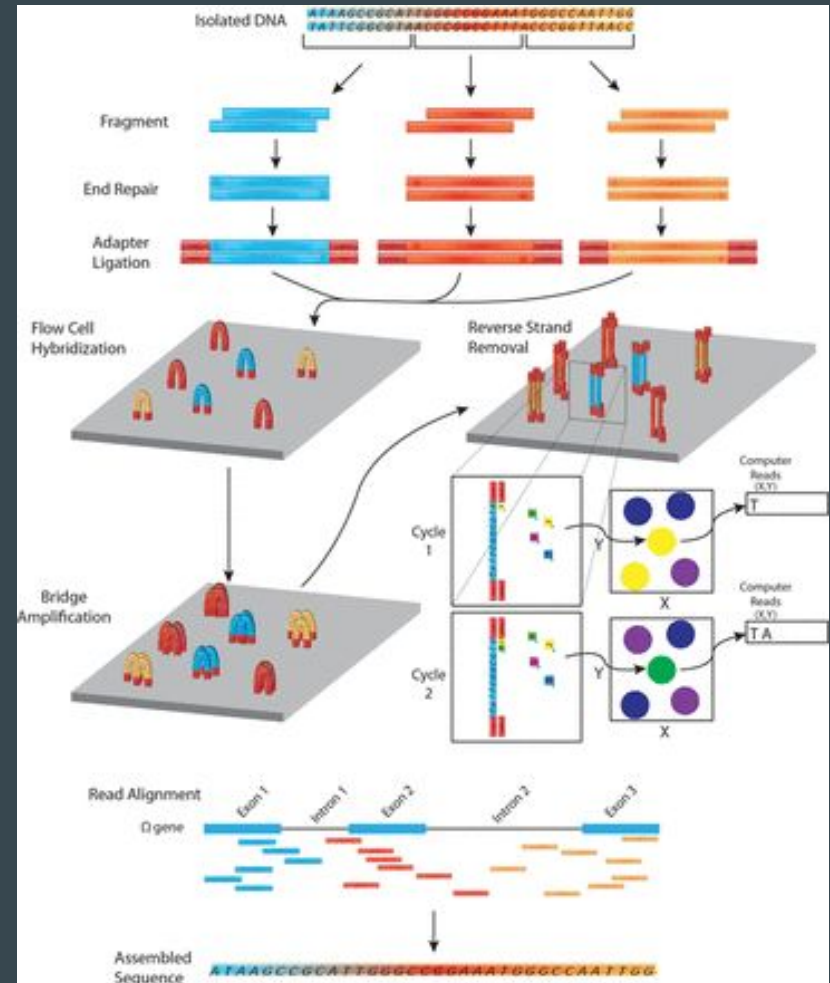
Up to 10,000 long reads

Up to 240 MB per run (MB = 1 million bases)

Illumina Sequencing process

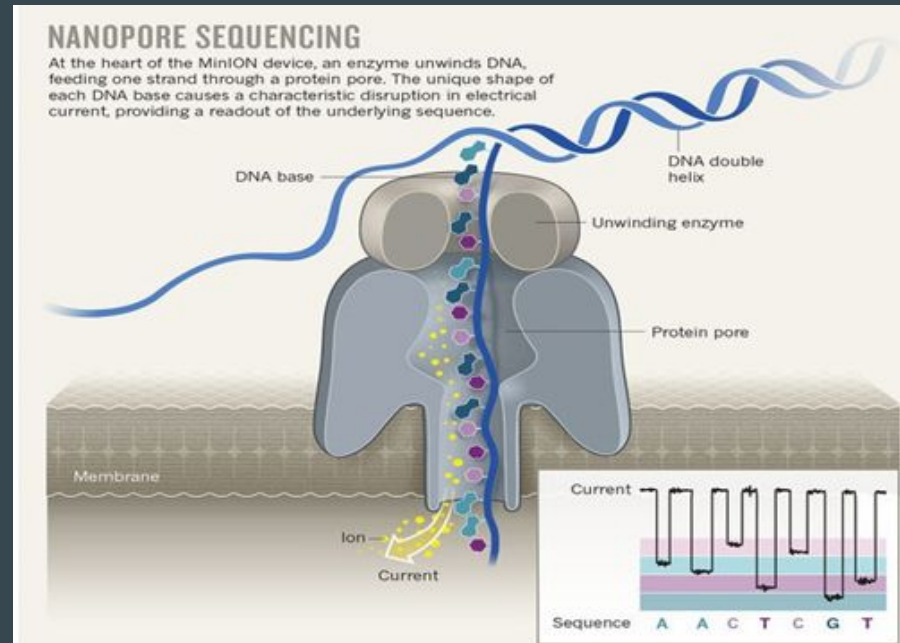
<https://www.youtube.com/watch?v=fCd6B5HRaZ8>

<https://emea.illumina.com/systems/sequencing-platforms/comparison-tool.html?langsel=/it/>



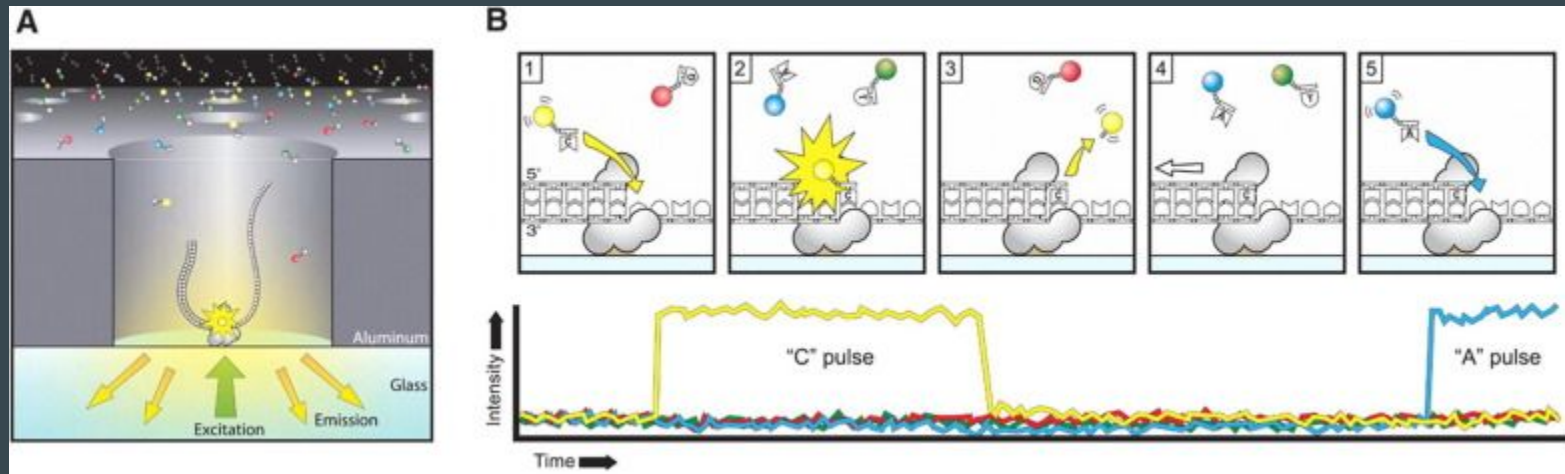
nanopore

<https://nanoporetech.com/applications/dna-nanopore-sequencing>



Pac Bio

<https://www.pacb.com/smrt-science/smrt-sequencing/>



Who we are

A Rapid Molecular Intervention

The Bio Arte is an innovative player in the field of **biomolecular research**, highly specialized on the **microbiota**.

In-depth knowledge, cutting-edge technology, and a team that is consistently up to date with the latest in **scientific research**: Those are the genes in our DNA.

The subject of our diagnostic analyses are all types of **Microorganisms**, ranging from those which are essential for our well-being to those which are pathogenic.

Follow us on: 



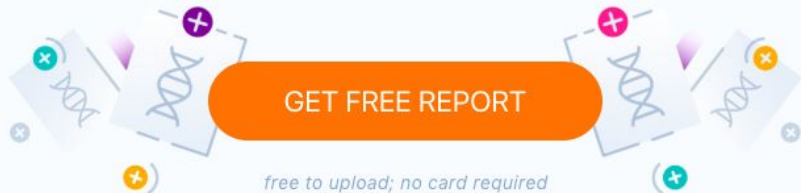
 CONTACT US

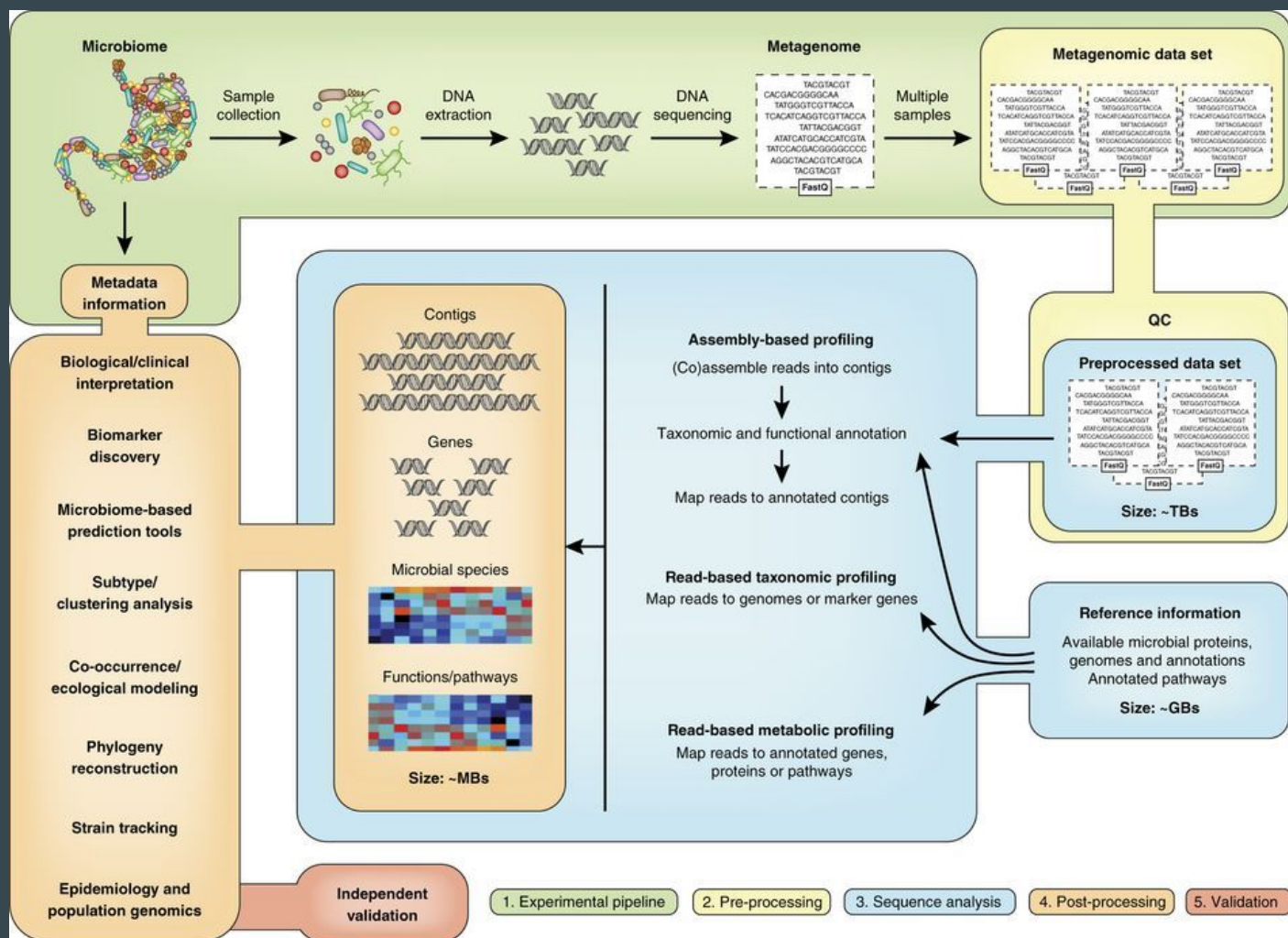


Decipher your raw DNA data

Go beyond genealogy — and learn how your DNA shapes your nutrition, fitness, personality and intelligence traits

Free additional analysis of your [23andMe](#), [Ancestry](#) or [MyHeritage](#) data





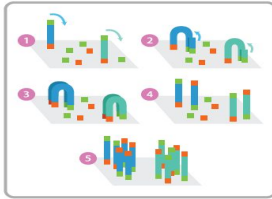
Next Generation Sequencing

1 Library Preparation

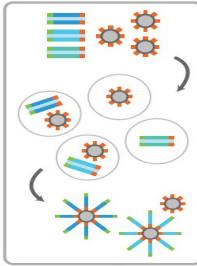


2 Clonal Amplification

Bridge PCR

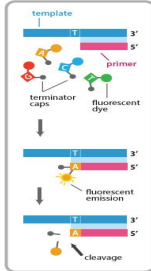


Emulsion PCR

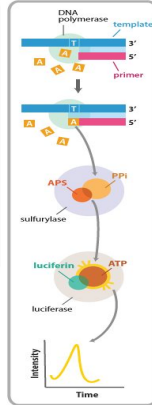


3 Cyclic Array Sequencing

Sequencing by Synthesis



Pyrosequencing



Sequencing by Ligation

